

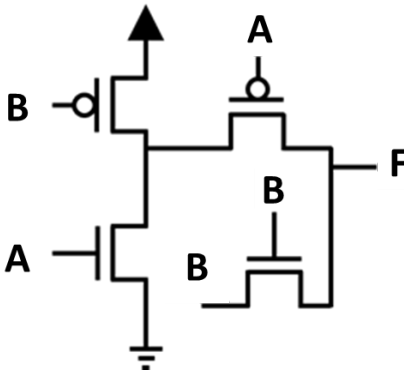
Semester: **Spring 2024**
Course Code: **CSE460**
Course Title: **VLSI Design**

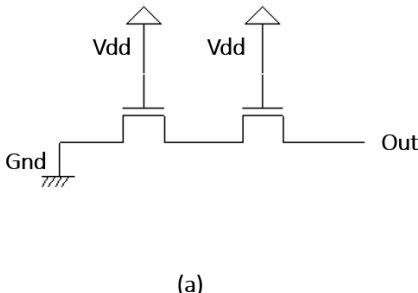
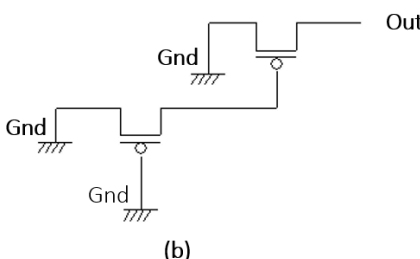
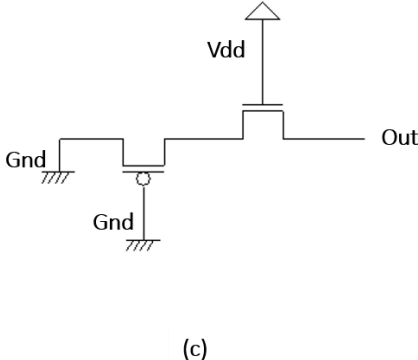
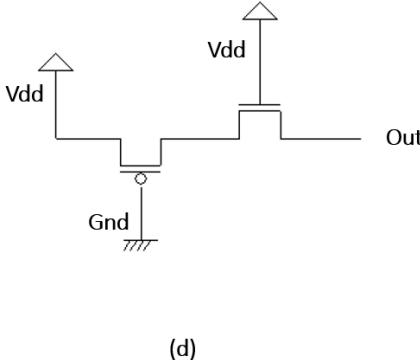
Midterm Exam
Full Marks: **15 x 3 = 45**
Time: **1 hour 15 mins**
Date: **8th Mar 2024**

Set A

Student ID:	Name:	Section:
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[**Question 4 is mandatory.** You may answer any **TWO** questions out of the remaining **THREE** along with **Question 4**. In total, you need to answer **THREE** questions out of **FOUR**.]

1. (CO1)	i. Complete the truth table in Figure 1b for the circuit shown in Figure 1a																	
			<table border="1" data-bbox="887 934 1216 1294"><thead><tr><th>A</th><th>B</th><th>F</th></tr></thead><tbody><tr><td>0</td><td>0</td><td></td></tr><tr><td>0</td><td>1</td><td></td></tr><tr><td>1</td><td>0</td><td></td></tr><tr><td>1</td><td>1</td><td></td></tr></tbody></table>	A	B	F	0	0		0	1		1	0		1	1	
	A	B	F															
	0	0																
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Figure 1a		Figure 1b																
(a)	Complete the truth table in Figure 1b for the circuit shown in Figure 1a .	4																
(b)	Explain your reasoning for the output F when A = 1, B = 0.	2																
ii. For a 0.8-μm process technology, tox = 15 nm , μ = 275 cm²/V.s , εox = (3.9)(8.85 × 10⁻¹⁴) F/cm and Vt = - 0.7 V .																		
(a)	Judging from the value of Vt and μ, comment on whether the MOSFET is NMOS or PMOS.	2																
(b)	Calculate Cox.	2																
(c)	For a MOSFET with aspect ratio W/L = 20 , calculate the values of β, Vsg and Vsd(min) needed to operate the transistor in the saturation region with a dc current of Id = 0.1 mA.	5																

2. (CO1)	(a) Design a CMOS compound gate that implements the following function: $Y = ((A \cdot B \cdot C) + D) \cdot E$	7	
	(b)  (a)  (b)  (c)  (d)	Determine the value of Out in all above conditions considering $V_{dd} = 3V$, $V_{tp} = -0.4V$, $V_{tn} = 0.5V$ and $Gnd = 0V$.	8
	3. (CO1)	Consider you need to design a temperature control system for your classroom at BRAC University. Your system has 1 Output D, and 3 inputs A, B and C that come directly from three different sensors installed at the room. When a certain threshold of the temperature is crossed, each sensor gives a logical HIGH output. Your logic circuit should turn on the AC (D = 1) when the majority of inputs are HIGH. [You can assume that you have the inputs available both in uncomplemented and complemented form]	
	(a) Write the truth table for the system.	5	
	(b) Determine the boolean logic function of the system using KMAP and design the circuit. How many MOSFETs do you require to implement your design? Can you design the circuit using NAND gates only? Show how	4+2 +2+ 2	

4. (CO2) **Fig 2a** shows that three 4-bit registers (**R1**, **R2**, and **R3**) are connected to a data bus. If the ‘in’ pins are high, the respective registers store the value from the data bus in the corresponding registers. Similarly, if the ‘out’ pins are high, the registers make the stored value available in the data bus. **R2** has an additional functionality of right shifting. If the **R_shift** pin is **high**, **R2** right shifts its content by **1 bit**. A control unit (**Fig 2b**) controls the registers and it generates the outputs sequentially as listed in **Table I**. The control unit completes the operation irrespective of the value of **w** once it’s triggered by **w = 1** initially.

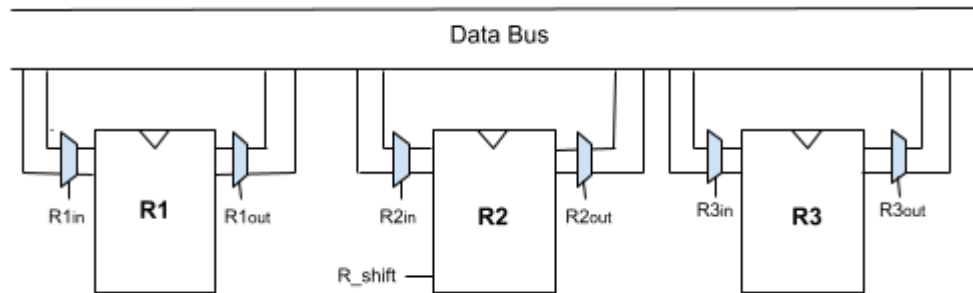


Fig 2a: Register arrangement

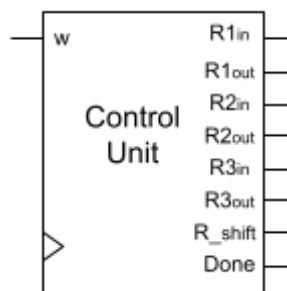


Fig 2b: Control Unit

Clock cycle	R1in	R1out	R2in	R2out	R3in	R3out	R_shift	Done
t1	0	1	1	0	0	0	0	0
t2	0	0	0	0	0	0	1	0
t3	0	0	0	1	1	0	0	1

Table I: Output values

- | | | |
|-----|--|----------|
| (a) | Assume the initial stored values of the registers as follows:
R1 = 0100, R2 = 0101, R3 = 1000.

What are the stored values of the registers after each clock cycle and briefly describe what’s happening? | 6 |
| (b) | Design the internal circuit of the ‘Control Unit’. You can use any type of state diagram and state assignment scheme to derive the circuit. | 9 |